

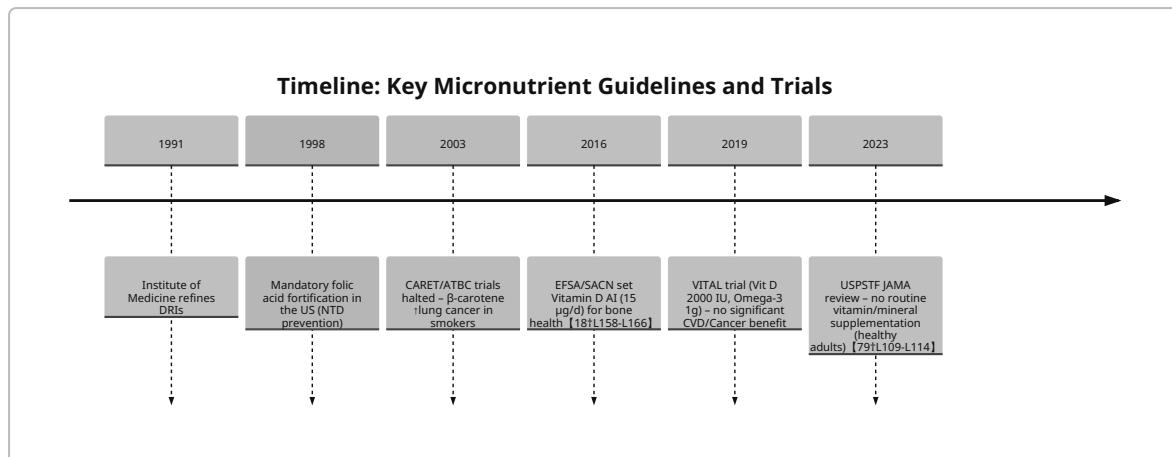
Executive Summary

Micronutrients (vitamins, minerals, trace elements, omega-3 fatty acids and related compounds) are essential for normal physiology. While most healthy people obtain sufficient micronutrients from diet, **deficiency remains common in vulnerable groups** (children and pregnant women in low-income countries, vegetarians/vegans, the elderly, certain ethnic groups and regions) [14†L106-L114]

[16†L144-L153]. Well-designed studies show that **general multivitamin/mineral supplements do not reduce chronic disease risk** in healthy populations [79†L109-L114]. However, **targeted supplementation** provides clear benefits in deficiency states or high-risk situations. For example, *periconceptional folic acid* (0.4–5 mg) (folate) **strongly prevents neural tube defects (NTDs)** [50†L136-L144]; *vitamin D* (10–20 µg/d) with calcium reduces rickets and osteomalacia, and modestly lowers fractures/falls in the elderly [18†L175-L183]; *vitamin B12* (cyanocobalamin/methylcobalamin) corrects deficiency anaemia and neuropathy in malabsorptive or vegan individuals [41†L120-L127]; and *iron* (30–60 mg elemental) prevents and treats iron-deficiency anaemia. By contrast, trials of high-dose antioxidants (β-carotene, vitamins E/C) in well-nourished adults showed **no benefit (and sometimes harm)** [79†L109-L114] [10†L33-L40].

This report reviews micronutrient definitions, physiological roles and global deficiency patterns by demographic group. It grades evidence (GRADE) for supplementation *beyond basic nutrition*, based on Cochrane reviews, systematic reviews and meta-analyses. Recommended intakes and therapeutic dosages are given (with authoritative sources). Safety issues (tolerable upper limits, adverse effects, drug/supplement interactions) are summarised. We compare bioavailability of different forms (e.g. methyl- vs cyanocobalamin for B12, heme vs non-heme iron). Synergistic co-factors and timing (e.g. vitamin C + iron, vitamin D + calcium/K2) are discussed. Population-specific guidelines (pregnancy, infancy, elderly, vegetarians etc.) and screening indications are outlined. A summary table compares nutrients by strength of evidence, doses and special considerations. Finally, we identify evidence gaps and research priorities (e.g. long-term RCTs in deficient populations, optimal dosing strategies).

Key recommendations: All women of childbearing age should take 400 µg folic acid (5 mg if high-risk) pre-conception through early pregnancy [50†L136-L144]. Infants and pregnant/lactating women require vitamin D (10–20 µg/d) year-round [18†L175-L183] [12†L71-L79]. Iron supplementation (30–60 mg/d) is advised in pregnancy to prevent anaemia [78†L109-L118]. B12 supplements (~2.4 µg/d, higher if malabsorption) are indicated for vegans/elderly [41†L120-L127]. Other supplements should be driven by deficiency testing or clear risk; routine high-dose antioxidants or excess fat-soluble vitamins are *not* recommended in well-nourished populations [79†L109-L114] [10†L33-L40]. Clinicians should individualize advice based on diet, risk factors and screening, using the quick-reference table below.



1. Definitions and Roles of Micronutrients

Micronutrients include **vitamins** (organic cofactors), **minerals/trace elements** (inorganic ions) and **essential fatty acids** (e.g. omega-3s) needed in small amounts for health. Each performs specific roles: e.g. *vitamin A* is key for vision and immunity; *vitamin D* regulates calcium/bone metabolism; *B-vitamins* (thiamine, riboflavin, niacin, B6, B12, folate, etc.) are coenzymes in energy production and DNA synthesis; *vitamin C* is an antioxidant and collagen cofactor; *vitamin E* protects cell membranes; *vitamin K* is required for blood clotting and bone matrix proteins. Minerals like *calcium* and *magnesium* maintain bone and muscle function, *iron* is essential for hemoglobin, *zinc* for immune/enzymatic function, *iodine* for thyroid hormones, and *selenium* for antioxidant enzymes. Essential omega-3 fatty acids (EPA/DHA) support cell membranes, cardiovascular health and fetal neurodevelopment. Most of these nutrients cannot be synthesized by the body and must be obtained from diet or supplements.

A balanced diet usually supplies adequate micronutrients. However, certain lifestyles (e.g. vegan diets), life stages (infancy, pregnancy, old age) or conditions (malabsorption, chronic disease) can increase requirements or reduce intake, leading to **deficiency**. Deficiency disrupts metabolic pathways and causes characteristic diseases (e.g. scurvy from vitamin C deficiency, pellagra from niacin deficiency, osteomalacia from vitamin D deficiency). Many micronutrients also have **upper intake limits** (ULs) beyond which toxicity or adverse effects can occur (see sections below).

2. Prevalence of Micronutrient Deficiencies

Micronutrient deficiencies are unevenly distributed globally and across subpopulations. Low-income countries carry the highest burden, but even affluent societies have pockets of insufficiency. Key patterns include:

- **Vitamin A:** Globally, **subclinical vitamin A deficiency** affects an estimated **30–40% of children under 5** in developing regions [14†L106-L114]. It is a major cause of preventable childhood blindness and increases infection severity. Prevalence is highest in sub-Saharan Africa and South/Southeast Asia, especially among preschool children and pregnant women [14†L106-L114].
- **Vitamin D:** Insufficiency is common worldwide. In Europe and North America, up to **20–30% of adults** have serum 25(OH)D below 25–30 nmol/L, rising to **50–80% in institutionalised or homebound elderly** [18†L175-L183]. Darker-skinned or veiled individuals, people at high latitude or indoor workers are at especially high risk.

- **B-vitamins: Folate (B9)** deficiency is seen in populations without folic acid fortification (e.g. some South Asian regions) and among women with poor diet or malabsorption. Deficiency during pregnancy leads to neural tube defects. **Vitamin B12** deficiency is common in older adults (due to atrophic gastritis or pernicious anemia) and strict vegans (no animal foods). In western countries, ~6% of adults over 60 have biochemically low B12 [41†L120-L127] . Suboptimal intakes of B6 and riboflavin occur in alcoholics and the elderly.
- **Vitamin C:** Full-blown scurvy is now rare, but marginal insufficiency may occur in **smokers, low-income populations, the elderly, or those with limited fresh produce intake** [54†L346-L354] .
- **Vitamin E/K:** Deficiencies are uncommon. Newborns are given vitamin K at birth to prevent hemorrhage.
- **Iron: Iron-deficiency anemia (IDA)** is the most common micronutrient deficiency globally. Anemia affects ~**40% of pregnant women** and ~**30% of all women of reproductive age** worldwide [16†L146-L154] . Children under 5 have ~40% anemia prevalence [16†L162-L166] . Much of this is due to iron deficiency (plus other causes). Risk groups include menstruating women, pregnant women, toddlers, and vegetarians.
- **Iodine:** Iodine deficiency persists in some mountainous/remote areas without iodized salt; historically, 30–40% of the world's population were at risk. Many countries now have iodization programs, yet mild deficiency (median urinary iodine <100 µg/L) remains in parts of Europe, Africa and Asia [45†L125-L134] . Pregnant women especially need adequate iodine (150–250 µg/d) to prevent cretinism and cognitive impairment in offspring.
- **Calcium/Magnesium/Zinc:** Diets low in dairy, vegetables or whole grains can lead to inadequate calcium or magnesium intake (often subclinical, potentially affecting bone or muscle health). Zinc deficiency (due to low-meat diets or malabsorption) is estimated in up to 50% of children in low-income countries, contributing to growth retardation and immune dysfunction.
- **Omega-3 Fatty Acids:** Many people do not consume the recommended amounts of EPA/DHA. However, there is no clinical “deficiency” syndrome (ALA insufficiency is rare in normal diets). Pregnant/lactating women often fall short of the ~250–300 mg/d DHA recommended for fetal development.

Population groups at high risk: Pregnant and lactating women have increased needs for folate, iron, iodine, vitamin D and calcium. Infants and young children require vitamin D (for rickets prevention) and iron (for rapid growth). Older adults (especially 65+) often need supplements of vitamin D (due to low sunlight exposure and skin synthesis) and vitamin B12 (poor absorption). Strict vegetarians/vegans require B12 (cobalamin) and possibly DHA/EPA supplements (unless eating DHA-enriched foods). People with malabsorptive disorders (celiac, IBD, bariatric surgery) may need extra vitamins A/D/E/K and B12 due to fat malabsorption. Low-income and food-insecure groups are at risk for multiple deficiencies, especially vitamins A, D, iron and iodine [14†L106-L114] [16†L146-L154] .

3. Evidence Strength for Supplementation (GRADE)

The efficacy of supplementation depends on the context. In **non-deficient, healthy individuals**, high-quality trials generally find minimal benefit from routine vitamin/mineral supplements in preventing chronic diseases [79†L109-L114] . For example, recent meta-analyses (GRADE: low to moderate) show

no significant reduction in cancer, cardiovascular events or all-cause mortality from multivitamins or single antioxidants in well-nourished adults [79†L109-L114] [10†L33-L40]. A US Preventive Services Task Force report concluded “little or no benefit” of supplementation for primary prevention, noting a small cancer incidence benefit only with multivitamins [79†L109-L114]. Harmful effects were observed for some high-dose antioxidants (e.g. β -carotene increased lung cancer in smokers, vitamin E slightly raised hemorrhagic stroke risk) [79†L109-L114]. Thus the **GRADE of evidence** is “high” for *no benefit* (null effect) of most supplements in healthy people for CVD/cancer outcomes [79†L109-L114].

In contrast, **targeted supplementation in deficiency or high-risk conditions** has strong evidence of benefit. For example:

- **Folate (Vitamin B9):** *High-quality evidence* (Cochrane, GRADE: high) shows that **preconceptional folic acid** (typically 0.4–5 mg daily) *dramatically* reduces neural tube defects [50†L136-L144]. A pooled meta-analysis (5 trials, 6708 births) found folic acid ≥ 0.36 mg/d cut NTD risk by ~69% (RR ≈ 0.31 , 95%CI 0.17–0.58) [50†L136-L144]. This is a paradigm of evidence-driven public health. No other micronutrient has similarly high-quality preventive evidence. (Folate also prevents megaloblastic anemia; however, excess folate can mask B12 deficiency – hence UL of 1000 μ g.)
- **Vitamin D (with Calcium):** The evidence is mixed but supportive for bone outcomes. Meta-analyses (GRADE: moderate) indicate that *combined* calcium (1000–1200 mg) plus vitamin D (800–2000 IU/d) **reduces fracture risk and falls** in elderly patients (especially institutionalised) more than either alone [18†L175-L183]. Vitamin D alone has little effect on fractures unless levels are very low. The US Endocrine Society and UK SACN endorse vitamin D 10–20 μ g/d for elderly/pregnant for bone health. Recent trials (e.g. VITAL) found no benefit for cancer/CVD prevention, so the GRADE for those outcomes is “low/insufficient.” In summary, the evidence is *moderate* for skeletal benefits (particularly combined with calcium) and *weak* for extraskelatal disease prevention.
- **Iron:** Iron supplementation (30–60 mg elemental per day) **clearly improves hemoglobin and iron stores** in iron-deficiency anaemia (IDA). Meta-analyses confirm high-grade evidence for correction of IDA with modest doses of oral iron. However, iron supplements do *not* prevent morbidity in non-anemic persons (GRADE low for prevention). In children, iron-deficiency treatment improves growth and cognitive outcomes modestly if started early.
- **Vitamin B12 (Cobalamin):** High-quality evidence supports supplementation for proven deficiency. In pernicious anemia or absorption disorders, high-dose oral or intramuscular B12 rapidly corrects anemia and neurological symptoms. In vegetarians with low B12, daily supplements (250–500 μ g) or sublingual forms are effective (GRADE high). There is little evidence to support B12 for cardiovascular or cognitive outcomes beyond repleting deficiency.
- **Vitamin A:** In areas of deficiency, high-dose vitamin A in children (100,000–200,000 IU semiannually) reduces mortality and diarrheal disease (Cochrane reviews: GRADE high for mortality reduction in VAD settings). For vision/night blindness, the benefit is clear. In well-nourished adults, no benefit of supplementation has been shown, and high dose can be toxic (bones, teratogenic).
- **Vitamin C:** Trials show **no reduction in CVD or cancer** (GRADE high: null result). Regular intakes of ~1 g/day (above RDA) modestly shorten cold duration by ~10% (GRADE moderate) but do not significantly prevent colds in general. Thus routine high-dose C is not supported in non-deficient people. It *does* reliably cure scurvy if intake ≥ 10 mg/d.

- **Vitamin E:** Large trials found **no benefit for cardiovascular or cancer outcomes**, and high doses (~400–800 IU/d) may slightly increase all-cause mortality or hemorrhagic stroke (GRADE high for no benefit, moderate for harm). Therefore supplements are not recommended for prevention in the general population 【79†L109-L114】 【10†L33-L40】 .
- **Omega-3 Fatty Acids (EPA/DHA):** Evidence is mixed. Meta-analyses (GRADE moderate) generally find **no significant reduction in major CVD events** with ~1 g/d supplements, except in specific subgroups. High-dose EPA (e.g. 4 g/d in REDUCE-IT) lowered events in hypertriglyceridemic patients. The AHA gives omega-3 a Class IIa recommendation (1 g/d) in secondary prevention 【64†L126-L130】 . In pregnancy, observational data link maternal DHA to neurodevelopment (GRADE low), but RCTs of DHA supplementation have inconsistent results. Overall, evidence is **insufficient** for broad preventive claims, though triglyceride lowering is consistent.
- **Zinc:** Therapeutic (≥ 75 –100 mg/d zinc acetate) has been shown to reduce duration of colds if begun within 24 h (GRADE moderate). Routine zinc supplementation in non-deficient adults has no clear benefit. In children in LMICs, zinc reduces diarrheal morbidity (GRADE high).
- **Iodine:** In iodine-deficient regions, supplementation (salt iodization, 150 µg/d in pregnancy) prevents goiter and cretinism (GRADE high evidence from decades of public health experience). Excess iodine, however, can induce hyperthyroidism in the elderly, so intake should be controlled (UL ~600 µg 【45†L125-L134】).

Other nutrients (e.g. selenium, magnesium, chromium) have some proposed benefits but evidence from high-quality trials is either negative or equivocal (GRADE low). For instance, selenium supplementation reduced cancer incidence in deficient regions but not in populations with adequate status.

Summary: The strength of evidence (GRADE) for **supplementation benefit** is high for targeted uses (folate for NTDs, iron for anemia, vitamin D/Ca for bone in deficiency, vitamin A in VAD areas), and low for broad health claims in well-nourished adults 【79†L109-L114】 【50†L136-L144】 .

4. Recommended Intakes and Therapeutic Dosages

Recommended Dietary Allowances (RDAs) or Adequate Intakes (AIs) and upper limits (ULs) vary by nutrient (see Table 1). We summarise key values (adult, non-pregnant):

- **Vitamin A:** RDA ~700 µg RAE (women), 900 µg (men). UL 3000 µg (preformed retinol) 【79†L109-L114】 . (Beta-carotene has no UL, but >15 mg may cause carotenoderma.)
- **Vitamin D:** RDA/AIs ~15 µg (600 IU) adults 【18†L158-L166】 (EFSA sets 15 µg). UL ~100 µg (4000 IU).
- **Vitamin E (α-tocopherol):** RDA 15 mg (22.4 IU natural form). UL ~1000 mg (1500 IU) 【67†L1-L4】 .
- **Vitamin K:** AI ~70 µg. No UL established (toxicity is rare).
- **Vitamin C:** RDA ~75 mg (women), 90 mg (men) 【55†L1-L4】 ; smokers +35 mg. UL 2000 mg (mostly GI upset) 【59†L7-L10】 .

- **Thiamine (B1):** RDA ~1.1–1.2 mg; UL not specified (water-soluble).
- **Riboflavin (B2):** RDA ~1.1–1.3 mg; UL none.
- **Niacin (B3):** RDA ~16 mg NE (men), 14 mg (women). UL ~35 mg (flushing with nicotinic acid).
- **Vitamin B6:** RDA ~1.3–1.7 mg; UL 25 mg (neuropathy) 【43†L7-L10】 .
- **Folate (B9):** RDA 400 µg DFE (dietary folate equivalents; ~400 µg folic acid). In pregnancy, 600 µg DFE. UL 1000 µg folic acid (prevents masking B12 deficiency) 【76†L31-L39】 .
- **Vitamin B12:** RDA ~2.4 µg 【41†L120-L127】 ; no UL (excess is excreted) 【43†L7-L10】 .
- **Calcium:** RDA 1000 mg (age 19–50) 【36†L119-L125】 ; UL 2500 mg.
- **Magnesium:** RDA ~310–420 mg; UL (supplements) 350 mg (to avoid diarrhea).
- **Iron:** RDA 8 mg (men), 18 mg (women 19–50) 【78†L109-L118】 ; UL 45 mg.
- **Zinc:** RDA 8 mg (women), 11 mg (men); UL 25 mg.
- **Iodine:** RDA 150 µg; UL 600 µg (risk of thyrotoxicosis).
- **Selenium:** RDA ~55–70 µg; UL ~300–400 µg (EU: 255 µg) 【3†L254-L262】 .
- **Omega-3 (EPA+DHA):** No formal RDA. Many guidelines suggest ~250–500 mg combined EPA+DHA per day for adults. In pregnancy/lactation, ~200–300 mg DHA.

These values come from authoritative bodies (EFSA, IOM/NAM, WHO) and are listed in Table 1 below. *Therapeutic dosages* can exceed RDAs: e.g. folate 4–5 mg for high-risk pregnancy, iron 60–180 mg for IDA, vitamin D 20–50 µg (800–2000 IU) for severe deficiency. Specific dosing should be guided by blood levels and clinical context.

5. Safety, Upper Limits and Adverse Effects

Most vitamins/minerals have wide safety margins when consumed near RDA. However, **hypervitaminosis** and toxicity can occur with high-dose supplements:

- **Fat-soluble vitamins (A, D, E, K)** accumulate in tissues. Excess vitamin A (especially preformed retinol) causes headache, teratogenicity, liver damage (UL 3000 µg). Vitamin D excess (hypercalcemia, vascular calcification) above UL. High vitamin E (>1000 mg) may increase bleeding risk (interferes with vitamin K) and mortality 【79†L109-L114】 .
- **Water-soluble vitamins** are generally excreted, but high doses can still cause harm. For example, vitamin B6 >200 mg can cause sensory neuropathy (UL 25 mg). Excess folic acid (>1000 µg) masks B12 deficiency 【76†L31-L39】 . Niacin (B3) >35 mg causes flushing; pharmacologic doses (1–3 g) can cause liver toxicity. High vitamin C (>2 g) may cause GI upset and kidney stones.

- **Minerals:** Calcium >2500 mg/d can cause kidney stones and interfere with zinc/iron absorption. Iron supplements often cause constipation, nausea, or GI bleeding at high doses. Zinc >40 mg can induce copper deficiency and anemia. Selenium above UL can cause selenosis (hair loss, neuropathy). Iodine excess (e.g. >1000 µg) may trigger thyroid autoimmunity or hyperthyroidism in susceptible individuals.
- **Omega-3s:** Generally safe; high doses (>3 g/d) may increase bleeding tendency and LDL in some individuals. Marine oils can contain contaminants (mercury, PCBs), so purified supplements are recommended.
- **Interactions:** Many micronutrients interact with medications or other supplements. For example, **vitamin K** antagonizes warfarin (coumarin) anticoagulants. **Calcium, magnesium and aluminum** reduce absorption of tetracyclines and fluoroquinolones. **Iron and calcium** compete for absorption with each other and with zinc. **Vitamin C** enhances non-heme iron absorption. **High-dose vitamin A/E** may potentiate anticoagulants, while vitamin D enhances vitamin K-dependent clotting factors. **B vitamins** can lower levels of some drugs (e.g. B6 with L-dopa). Clinicians should review patient medications when recommending supplements.

Supplement purity and quality can also impact safety: look for third-party tested products to avoid contaminants or incorrect dosing.

6. Bioavailability and Preferred Forms

The **chemical form** of a supplement can affect its absorption:

- **B12 (Cobalamin):** Available as cyanocobalamin or methylcobalamin (also adenosyl- and hydroxycobalamin). Methylcobalamin is the naturally active form, but cyanocobalamin is stable and effective after conversion. Studies show *no major difference in clinical efficacy or absorption* between forms 【42†L1-L4】. In malabsorption (pernicious anemia), high-dose oral cyanocobalamin or intramuscular injections are both effective.
- **Folate:** Commonly as folic acid (synthetic, in fortified foods/supplements) or 5-methyltetrahydrofolate (5-MTHF). 5-MTHF is already the active metabolite and may be better absorbed in people with MTHFR polymorphisms, but in practice folic acid is effective for most. For pregnancy, folic acid (0.4–5 mg) remains standard 【50†L136-L144】.
- **Iron:** *Heme iron* (from animal sources, e.g. heme polypeptide or heme iron tablets) is absorbed about **2–3 times** better than elemental (non-heme) iron 【78†L109-L118】. Among non-heme iron salts, **ferrous fumarate** or **ferrous sulfate** are more bioavailable than ferric forms. *Iron bisglycinate* (chelate) is also well absorbed with fewer GI side effects. Taking iron **with vitamin C** (ascorbic acid) markedly increases absorption, while **calcium, tea/coffee and phytates** inhibit it.
- **Calcium:** Calcium carbonate requires stomach acid (take with meals); citrate (or as gluconate) is better absorbed when gastric pH is low (elderly or on PPIs), albeit at higher cost.
- **Magnesium:** Mg oxide has lower bioavailability (~4%); magnesium citrate, gluconate or aspartate are better absorbed but more expensive.
- **Zinc:** Chelated forms (e.g. zinc picolinate) may absorb slightly better than sulfate, but overall differences are minor.

- **Vitamin D:** Cholecalciferol (D3, from animal/lanolin sources) raises serum 25(OH)D more effectively than ergocalciferol (D2, from plants) 【18†L175-L183】. Supplements should be taken with a fat-containing meal for optimal absorption.
- **Vitamin K:** Phylloquinone (K1) from plants vs menaquinones (K2) from fermented foods. K2 has longer half-life in blood. Some believe K2 (menaquinone-7) is preferable for bone/vascular health, but evidence is not definitive.
- **Omega-3:** Triglyceride (TG) forms (fish oil) vs ethyl esters (some purified products) vs free fatty acids. TG form is better absorbed. Algal DHA supplements are an alternative for vegetarians.

7. Timing and Cofactors

Optimizing absorption and synergistic effects often requires **timing and pairing**:

- **Vitamin C + Iron:** Take iron supplements with a glass of orange juice (vitamin C) to increase non-heme iron uptake. Conversely, avoid co-administering iron with calcium supplements or dairy.
- **Vitamin D + Calcium:** Co-supplementing vitamin D and calcium (as in many bone-health formulas) is more effective than either alone for bone density and fall prevention. Vitamin D ensures adequate calcium absorption, while calcium stimulates 25(OH)D activation in kidneys.
- **Vitamin D + Vitamin K2:** Some evidence suggests K2 (MK-7) may help direct calcium into bone rather than vessels, complementing vitamin D. However, this synergy lacks high-grade evidence; still, combined D+K2 products are common (no known harm with physiologic doses).
- **B vitamins:** Many B vitamins are water-soluble and can be taken together. Some complex supplements combine B6, B12, folate for homocysteine lowering, but RCTs have not shown CV benefit beyond homocysteine reduction.
- **Fat-soluble vitamins:** Should be taken with meals containing fat for absorption (e.g. vitamin D/E/K).
- **Evening vs morning:** B vitamins can be stimulating; taking B complex earlier in the day may reduce insomnia. Minerals (calcium, magnesium) can be taken in divided doses; magnesium at bedtime may help sleep/cramp.
- **Interactions with food:** As above, avoid iron with tea/coffee, antibiotics, or PPIs (which reduce stomach acid). Calcium can bind tetracyclines. High-dose zinc or iron should be spaced from other minerals to prevent absorption competition.

8. Population-Specific Recommendations and Screening

- **Pregnancy:** Universal prenatal vitamins (with folic acid 400–600 µg, iron 30–60 mg, iodine 150–250 µg, vitamin D 10 µg) are recommended 【12†L28-L32】 【78†L109-L118】. In the UK, all pregnant women are advised 400 IU (10 µg) vitamin D daily 【12†L71-L79】, and folic acid 400 µg from preconception through 12 weeks (5 mg if high risk) 【12†L28-L32】. Iron is supplemented if anemia is detected; otherwise intake of ≥30 mg/d plus diet is advised 【78†L109-L118】.

Screening: check hemoglobin and ferritin; measure B12/folate only if macrocytic anemia or risk factors.

- **Infants & Children:** Breastfed infants need 400 IU vitamin D from birth; formula-fed infants require vitamin D if intake <1000 mL formula/d. From age 6 months, introduce iron-rich solids; screening of hemoglobin at 1 year is common. Some countries fortify baby foods with iron. Universal infant hydroxocobalamin injections often cover B12 if maternal levels are adequate; if the mother is vegan or B12-deficient, infant B12 supplements are needed. For growth and immunity, children 1–4 years need adequate vitamin A and zinc (ensure dietary intake or supplements if diet poor).
- **Elderly (≥ 65 years):** Routine vitamin D (800–1000 IU) and calcium (1200 mg/d) is often recommended to maintain bone health, especially in institutionalized or nursing-home residents [18†L175-L183]. B12 (up to 1000 µg weekly oral or monthly injection) should be given if serum levels are borderline or intrinsic factor is low. Folate 400 µg/d is generally adequate unless atrophic gastritis present. Screening: measure vitamin D (25(OH)D) in falls/fracture patients; check B12 if cognitive/macrocytic anemia concerns.
- **Vegetarians/Vegans:** Ensure B12 intake (2.4 µg/d, or higher oral doses 100–500 µg). Calcium and vitamin D can be low without dairy – consider fortified foods or supplements (D3 as lichen or D2) especially in winter. Iron absorption from plant foods is lower (non-heme); pair with vitamin C and consider low-dose iron if anemic. Omega-3 (ALA) is abundant in plants, but EPA/DHA are lacking – algae-based DHA (250 mg/d) is advisable.
- **High-Physique/Endurance Athletes:** Generally meet needs via diet, but may consider magnesium, iron (if menstruating female athlete), and antioxidants (though high-dose antioxidant supplements are *not* recommended as they may blunt training adaptations).
- **Chronic Illness/Medications:** Certain drugs require supplements: e.g., metformin and PPIs can deplete B12; loop diuretics can cause magnesium loss; anticonvulsants interfere with vitamin D; methotrexate antagonizes folate (supplement 1 mg folic/folate). Clinicians should screen electrolytes/micronutrients based on long-term medication use.
- **Low-Income Settings:** WHO/UNICEF promote cost-effective supplementation: vitamin A capsules in children (every 6 months) to reduce mortality, iron/folic acid during pregnancy to reduce anaemia, zinc for acute diarrhea treatment, and salt iodization for whole population iodine sufficiency.

9. Cost-Effectiveness and Practical Guidance

Routine screening of all adults for micronutrient levels is generally *not* cost-effective in developed settings due to low pretest probability and the expense of assays. Instead, a **targeted approach** is advised: test and treat only those with suggestive symptoms or risk factors. For example, measure serum 25(OH)D if bone disease or malabsorption, check ferritin if anemia, or serum B12 if neuropathy or vegan diet. Empiric supplementation (at RDA doses) is reasonable in at-risk groups (e.g. pregnant women take folic acid/iron without testing).

Food fortification (e.g. folic acid in flour, iodine in salt, vitamin D in milk) is highly cost-effective population-wide. Standalone supplements should be recommended judiciously. Quality matters: advise patients to choose reputable brands with regulated dosing.

Clinicians should counsel patients that “megadoses” of vitamins usually have no extra benefit and can be harmful. Encourage a food-first approach – e.g. oily fish 2x/week for omega-3, leafy greens for folate, dairy or fortified plant milks for calcium/vitamin D – and use supplements to fill gaps. When prescribing supplements, specify the dose, form, and duration, and discuss possible side effects (e.g. iron’s GI upset).

10. Gaps in Evidence and Research Priorities

Important gaps include: long-term RCTs of supplementation in deficient subpopulations (e.g. vitamin D in truly deficient elderly for non-skeletal outcomes), comparative effectiveness of different nutrient forms (e.g. methylfolate vs folic acid), and optimal co-supplementation strategies (vitamin D + K2?). The cognitive effects of B vitamins, benefits of omega-3 in dementia, and genotypic influences on nutrient needs (nutrigenomics) remain under study. More data on cost-effectiveness in varied healthcare systems are needed.

Overall, future research should focus on **precision nutrition**: identifying who truly benefits from which supplement, rather than blanket recommendations. Monitoring emerging evidence from high-quality trials (GRADE, Cochrane) will remain essential.

11. Tables

Table 1. Summary of Key Micronutrients: Roles, RDA/AI, UL, Best Forms, Interactions, Target Groups.

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Target Groups
Vitamin A (retinol)	Vision, immunity; prevents xerophthalmia (high-dose supplements reduce mortality in deficient children, GRADE high) 【14†L106-L114】 . <i>No benefit in well-nourished adults.</i>	700 µg RAE (women); 900 µg (men) 【79†L109-L114】	3000 µg (preformed)	Retinol (preformed) vs β-carotene (provitamin A)	High-fat meals ↑ absorption; orlistat/bile-binding resins ↓ absorption.	Prevalence of deficiency is high in populations with low intake of animal products, including vegetarians and vegans. Deficiency is also common in people with malabsorption syndromes, liver disease, and alcoholism.

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Ta
Vitamin D (cholecalciferol)	Calcium homeostasis, bone health (prevents rickets/osteomalacia; VitD+Ca cuts fractures in elderly, GRADE moderate) 【18†L175-L183】 . No proven CVD/Cancer benefit (GRADE low).	15 µg (600 IU) (AI for EU by EFSA 【18†L158-L166】); 10 µg (400 IU) in UK for at-risk.	100 µg (4000 IU)	D3 (cholecalciferol) better than D2; take with fat	Fatty foods ↑ absorption. Some drugs (steroids, anticonvulsants) ↓ VitD.	EL pr in su 20 25 os fa
Vitamin E (α-tocopherol)	Antioxidant; no proven benefit for CVD/Cancer (GRADE high – trials null) 【79†L109-L114】 ; high-dose may ↑ hemorrhagic stroke risk.	15 mg α-tocopherol (≈22 IU) 【66†L140-L149】	1000 mg (1500 IU) 【67†L1-L4】	Natural RRR-α-tocopherol (d-α) has higher activity than dl-α.	High-dose E antagonizes vitamin K (↑ bleeding), ↑ Warfarin effect.	D ra su or de de (n
Vitamin K (phyloquinone/menaquinone)	Clotting factors, bone proteins. Few trials; no routine supplement needed except newborns.	AI ~70 µg (UK/EU)	None established (no toxicity)	K1 (leafy greens); K2 (menaquinones in natto/cheese)	Broad-spectrum antibiotics ↓ gut K2. Warfarin: vitamin K reverses effect.	N si K w co in
Vitamin C (ascorbate)	Antioxidant, collagen synthesis; prevents scurvy (deficiency disease). RCTs: no CVD/Cancer benefit; modest ↓ cold duration (8–14%, GRADE moderate) 【80†L0-L3】 .	75 mg (women), 90 mg (men) 【55†L1-L4】	2000 mg 【59†L7-L10】	Ascorbic acid (water-soluble); liposomal or Na-salt form for GI tolerance.	↑ Iron absorption. ↓ Aluminum absorption. High doses can ↑ oxalate kidney stones.	Sn +3 M or pa ne

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Ta N
Thiamin (B1)	Carbohydrate metabolism; nerve function. Deficiency (beriberi) in alcoholism/malnutrition.	1.1 mg (women), 1.2 mg (men)	None known	Thiamine mononitrate (stable in foods).	DFP (iron chelator) ↑ thiamine excretion.	Al di ↑ de ne he (w
Riboflavin (B2)	Electron transport (FAD), redox reactions. Deficiency causes cheilosis, corneal keratitis.	1.1 mg (women), 1.3 mg (men)	None known	Riboflavin (FMN cofactor); high-dose (supplements) are safe.	Some drugs (e.g. tricyclics) may ↓ riboflavin.	Ra ve pe ph m
Niacin (B3)	NAD/NADP cofactors in metabolism. Deficiency → pellagra (dermatitis, diarrhea, dementia). Therapeutic (1–3 g/d) ↑ HDL, but causes flushing (Grade A evidence for dyslipidemia).	14 mg NE (women), 16 mg (men)	35 mg (flush)	Nicotinic acid (flush-causing) vs nicotinamide (no flush).	Alcohol ↑ risk of hepatotoxicity with niacin; statins may ↑ side effects.	Pe W de ur m
Vitamin B6 (pyridoxine)	Cofactor in AA metabolism. Deficiency: neuropathy, sideroblastic anemia. Used (sometimes) for PMS, carpal tunnel (evidence limited).	1.3–1.7 mg (adults)	25 mg	Pyridoxine hydrochloride in supplements; active form = P-5-P (pyridoxal phosphate).	Isoniazid and penicillamine ↑ B6 requirement.	H (≥ sp an B m ne at

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Ta N
Folate (B9)	DNA synthesis, homocysteine metabolism; prevents megaloblastic anemia. <i>Strong evidence (GRADE high)</i> : 0.4–5 mg folic acid prevents NTDs 【50†L136-L144】 .	400 µg DFE (women), 400 µg (men); 600 µg in pregnancy 【50†L136-L144】	1000 µg (folic acid) 【76†L25-L34】	Folic acid (synthetic) or methylfolate (active).	Methotrexate antagonizes folate (supplement if on MTX).	Al ch ag ac if or ep 【 L1 M an su de
Vitamin B12 (cobalamin)	DNA synthesis, myelin formation. Deficiency → megaloblastic anemia, neuropathy (especially in elderly, vegans).	2.4 µg (RDA) 【41†L120-L127】	None (no toxicity) 【43†L7-L10】	Cyanocobalamin (stable, common); methylcobalamin (active form). Injectable hydroxycobalamin in some countries.	PPIs/H2-blockers and metformin ↓ B12 absorption. Chloramphenicol blunts anemia response.	Ve el ne 50 10 in M m ac bo
Calcium	Bone/mineral structure, muscle/nerve function. RDA based on bone accretion – ~1000 mg/day 【36†L119-L125】 . Calcium+VitD modestly ↓ osteoporotic fractures.	1000 mg (adults) 【36†L119-L125】 ; 1200 mg (women >50, men >70)	2500 mg	Calcium carbonate (best with food, needs acid) vs citrate (acid-independent).	↓ Iron and Zn absorption if co-ingested. Thiazide diuretics ↑ serum Ca.	Pe w el 10 Ca da di su

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Ta N
Magnesium	Over 300 enzymes (ATP metabolism, muscle relaxation). Deficiency: cramping, arrhythmias (rare from diet, may occur with diuretics).	310–420 mg	350 mg (supplements only; GI limit)	Magnesium citrate/malate = higher bioavailability; oxide = cheaper, lower absorption.	Competes with Ca for absorption. Aminoglycosides loop diuretics ↑ excretion.	Cl di m ca Co su le (w
Iron	Hemoglobin/myoglobin, O ₂ transport. RDA 8–18 mg (women higher). Evidence: oral iron 30–60 mg/d corrects IDA. No benefit in non-anemic (GRADE moderate).	8 mg (men), 18 mg (women)	45 mg	Ferrous sulfate/fumarate/gluconate (choose for GI tolerance). Heme iron polypeptide (better absorption).	↑ Fe by vitamin C; ↓ Fe by Ca, tea, phytates, PPIs. Tetracyclines, quinolones ↓ absorption.	W ch an su 60 I L1 an ir to
Zinc	Immune function, wound healing, cofactor. Supplementation (20–75 mg/d) shortens cold duration if started <24 h (GRADE moderate).	8 mg (women), 11 mg (men)	40–45 mg (adults)	Zinc acetate or gluconate lozenges (for colds), zinc sulfate or gluconate tablets.	High Ca/Fe supplements interfere; excess zinc → Cu deficiency.	Ve (h di al de lo m co
Iodine	Thyroid hormones (T4/T3 synthesis). Prevents goiter/cretinism.	150 µg; 250 µg (pregnancy)	600 µg	Iodized salt, potassium iodide/citrate.	Excess iodine may precipitate thyrotoxicosis; sushi/seaweed may provide very high iodine.	Pr la ≥ (ic su M io en an

Nutrient	Main Roles & Evidence (GRADE)	RDA/Adequate Intake (Adults)	Tolerable UL	Preferred Forms	Key Interactions*	Ta N
Selenium	Antioxidant enzymes (glutathione peroxidases), thyroid metabolism.	~55–70 µg	~300 µg (US); 255 µg (EFSA)	Selenomethionine (food form) or selenite/selenate.	High sulfur amino acids (garlic) may ↑ excretion.	D Ke ris Ch su ~5 H (> ↑
Omega-3 (EPA+DHA)	Membrane fluidity, eicosanoid metabolism. Lowers TGs (strong RCT evidence). Mixed results for CVD prevention [64†L120-L128] .	AI: ALA: 1.1–1.6 g; no RDA for EPA/DHA. 250–300 mg EPA+DHA often recommended.	None established (excess bleeding risk at >3 g/d)	Fish oil (TG form); algal oil (DHA) for vegetarians.	Vitamin E may be depleted by oxidized fish oil. High doses prolong bleeding time.	Pa co di EF (A w ≥ fo Ve su D

**Interactions*: minerals/vitamins often compete for absorption or metabolism as noted above. Patients should be counselled on spacing or combining based on the nutrient._

Table 2. Quick Reference for Clinicians (Selected Supplements) – one-page summary table for easy clinical use.

Nutrient	Dose (Adults)	UL / Caution	Form to Use	Key Interaction / Note	When to Supplement
Folic Acid	400 µg/d (all women of childbearing age); 5 mg/d if high-risk	1000 µg (mask B12 def)	Folic acid tablets	MTX (folate rescue)	All women preconception to 12 wks [50†L136-L144] ; macrocytic anemia
Iron	30–60 mg Fe/d (elemental)	45 mg/d (GI side effects)	Ferrous sulfate or fumarate	↑ by vit C, ↓ by Ca/ coffee	Iron-deficiency anemia (confirm ferritin), pregnancy if anemic [78†L109-L118]

Nutrient	Dose (Adults)	UL / Caution	Form to Use	Key Interaction / Note	When to Supplement
Vitamin D	10–20 µg/d (400–800 IU)	100 µg (4000 IU)	Cholecalciferol (D3)	Take with meal (fat)	Elderly, dark-skinned, institutionalized; osteoporosis prevention
Calcium	1000–1200 mg/d	2500 mg/d	Citrate (PM), carbonate (with food)	Ca ↓ Fe/Zn absorption	Postmenopausal bone health (with vit D)
Vitamin B12	2.4 µg/d (≥1000 µg oral daily or 1000 µg IM monthly if malabsorption)	None (water-soluble)	Cyanocobalamin or methylcobalamin	PPIs/metformin ↓ levels	Vegans, pernicious anemia, neurologic sx
Vitamin C	75–90 mg/d	2000 mg/d	Ascorbic acid	↑ Iron absorption, ↓ Al absorption	Scurvy prevention (low-dose); high-dose colds (~1g/d)
Zinc	8–11 mg/d	40–45 mg/d	Gluconate or acetate	↓ Cu absorption, ↓ ↑ Fe/Ca abs.	Shortens cold (≥75 mg/d early); deficiency in malnutrition
Vitamin K	70 µg/d	None	Phylloquinone (K1) or MK-7 (K2)	Antagonizes warfarin	Newborn IM injection; monitor in liver disease
Omega-3 (EPA+DHA)	250–500 mg EPA+DHA (1 g for secondary prevention)	>3000 mg (bleeding)	Fish oil capsules; algal oil	None major (thin blood)	Coronary disease (1g/d AHA), hypertriglyceridemia, pregnancy (200 mg DHA)

Notes: Doses are for adults; adjust for age/sex as per Table 1. UL = tolerable upper intake level. Forms: take supplements with food unless noted. Interaction/Note highlights major drug or nutrient interactions. "When to supplement" gives primary indications (other uses may exist).